

12 – JSON

A simple (Javascript) notation to describe complicated data

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YES / NO



JA / NEJ



OUI / NON

OUTLINE

- 1) JSON
- 2) Notation
- 3) Examples

1) JSON

- **JSON (JavaScript Object Notation)** is a way to write objects in Javascript
- JSON is now used by **MANY** programming languages because of its **power** and **simplicity**

“JSON (JavaScript Object Notation, pronounced /'dʒeɪsən/; also /'dʒeɪ,sn/) is an open standard file format and data interchange format that uses human-readable text to store and transmit data objects consisting of attribute–value pairs and arrays (or other serializable values). It is a common data format with diverse uses in electronic data interchange, including that of web applications with servers.” from <https://en.wikipedia.org/wiki/JSON>

2) NOTATION

- All JSON is on this slide!

{ and } mean "object"

[and] mean "array"

"key": "value" is a pair consisting of a key associated to a value

, is the separator between objects AND between pairs

2) NOTATION

- In JSON, ONLY 7 characters are used to define the structure of data:

{ } object

[] array

"key": "value" pair

, separator

3) EXAMPLES

```
{  
  "index": "0",  
  "first-name": "Jérôme",  
  "last-name": "Landré",  
  "jobStartDate": "2022-08-01",  
  "job": "Teacher",  
  "employer": "Jönköping University"  
}
```

3) EXAMPLES

```
{ "index": "0", "first-name": "Jérôme", "last-name": "Landré", "birth-date": "1972-10-12",  
  "job": "Teacher", "employer": "Jönköping University" }
```


3) EXAMPLES

```
{  
  "index": "0",  
  "first-name": "Jérôme",  
  "last-name": "Landré",  
  "jobStartDate": "2022-08-01",  
  "job": "Teacher",  
  "employers": [  
    "Jönköping University",  
    "Université de Reims Champagne-Ardenne",  
    "Université de Bourgogne"  
  ]  
}
```

3) EXAMPLES

```
{  
  "index": "0",  
  "first-name": "Jérôme",  
  "last-name": "Landré",  
  "jobStartDate": "2022-08-01",  
  "job": "Teacher",  
  "employers": [  
    { "id": "3", "name": "Jönköping University" },  
    { "id": "2", "name": "Université de Reims Champagne-Ardenne" },  
    { "id": "1", "name": "Université de Bourgogne" }  
  ]  
}
```

3) EXAMPLES

```
{ "employees": [  
  { "index": "123", "first-name": "Jérôme", "last-name": "Landré", "jobStartDate": "2022-08-01", "job": "Teacher",  
    "employers": [ { "id": "3", "name": "Jönköping University" },  
                   { "id": "2", "name": "Université de Reims Champagne-Ardenne" },  
                   { "id": "1", "name": "Université de Bourgogne" } ]  
  },  
  { "index": "251", "first-name": "Björn", "last-name": "Larsen", "jobStartDate": "2022-06-01", "job": "Teacher",  
    "employers": [ { "id": "2", "name": "Jönköping University" },  
                   { "id": "1", "name": "University of Skövde" } ]  
  },  
  { "index": "267", "first-name": "Bilal", "last-name": "Zafar", "jobStartDate": "2023-06-01", "job": "Teacher",  
    "employers": [ { "id": "2", "name": "Jönköping University" },  
                   { "id": "1", "name": "University of Stuttgart" } ]  
  }  
]
```

JSON GENERATOR

Generate

Reset



Help Ukraine!

Try the next version!

Help

Feedback

```
1  [
2    '{repeat(5, 7)}',
3    {
4      _id: '{{objectId()}}',
5      index: '{{index()}}',
6      guid: '{{guid()}}',
7      isActive: '{{bool()}}',
8      balance: '{{floating(1000, 4000, 2, "$0.00")}}',
9      picture: "http://placeholder.it/32x32",
10     age: '{{integer(20, 40)}}',
11     eyeColor: '{{random("blue", "brown", "green")}}',
12     name: '{{firstName()}} {{surname()}}',
13     gender: '{{gender()}}',
14     company: '{{company().toUpperCase()}}',
15     email: '{{email()}}',
16     phone: '+1 {{phone()}}',
17     address: '{{integer(100, 999)}} {{street()}} {{city()}} {{state()}} {{integer(100, 10000)}}',
18     about: '{{lorem(1, "paragraphs")}}',
19     registered: '{{date(new Date(2014, 0, 1), new Date(), "YYYY-MM-ddThh:mm:ss Z")}}',
20     latitude: '{{floating(-90.000001, 90)}}',
21     longitude: '{{floating(-180.000001, 180)}}',
22     tags: [
23       '{{repeat(7)}}',
24       '{{lorem(1, "words")}}'
25     ],
26     friends: [
27       '{{repeat(3)}}',
28       {
29         id: '{{index()}}',
30         name: '{{firstName()}} {{surname()}}'
31       }
32     ],
33     greeting: function (tags) {
34       return 'Hello, ' + this.name + '! You have ' + tags.integer(1, 10) + ' unread messages.';
35     },
36     favoriteFruit: function (tags) {
37       var fruits = ['apple', 'banana', 'strawberry'];
38       return fruits[tags.integer(0, fruits.length - 1)];
39     }
40   }
41 ]
```

```
1  [
2    {
3      "_id": "65085b8e8c3d0e69d1d002b9",
4      "index": 0,
5      "guid": "230626b6-0436-49f1-8451-e9992a6b33c6",
6      "isActive": false,
7      "balance": "$3,572.99",
8      "picture": "http://placeholder.it/32x32",
9      "age": 39,
10     "eyeColor": "blue",
11     "name": "Massey Atkins",
12     "gender": "male",
13     "company": "ZOINAGE",
14     "email": "masseyatkins@zoinage.com",
15     "phone": "+1 (879) 530-2845",
16     "address": "932 Baughman Place, Sultana, Oregon, 3707",
17     "about": "Voluptate labore laborum nulla duis sit dolor tempor aute. Velit voluptate consectetur ea sint sint proident amet. Elit veniam eiusmod ullamco sit veniam dolore aliqua non.\r\n",
18     "registered": "2014-05-30T06:32:50 -02:00",
19     "latitude": 23.201217,
20     "longitude": -51.549228,
21     "tags": [
22       "mollit",
23       "occaecat",
24       "exercitation",
25       "velit",
26       "ex",
27       "labore",
28       "in"
29     ],
30     "friends": [
31       {
32         "id": 0,
33         "name": "Kidd Castaneda"
34       },
35       {
36         "id": 1,
37         "name": "Lynne Rice"
38       },
39       {
40         "id": 2,
41         "name": "Aguirre Zimmerman"
42       }
43     ],
44     "greeting": "Hello, Massey Atkins! You have 3 unread messages.",
45     "favoriteFruit": "banana"
46   },
47   {
48     "_id": "65085b8e4634f7f7313d7d46",
49     "index": 1,
50     "guid": "e1bb30d2-0ef7-4476-b3a9-ca68d910a06a",
51     "isActive": true,
52     "balance": "$3,347.55",
53     "picture": "http://placeholder.it/32x32",
54     "age": 26,
55     "eyeColor": "brown",
56     "name": "Anderson Vargas",
57     "gender": "male",
58     "company": "CONFERIA",
59     "email": "andersonvargas@conferia.com",
60     "phone": "+1 (965) 427-2800",
61     "address": "333 Hubbard Place, Dellview, Mississippi, 9846",
62     "about": "Sunt minim duis deserunt reprehenderit. Ullamco laborum duis voluptate id sint veniam incidunt est deserunt occaecat cupidatat. In nostrud do anim reprehenderit reprehenderit eiusmod. Consequat quis commodo ad exercitation est dolore labore commodo. Fugiat anim sint consequat anim tempor ad reprehenderit ea ad enim adipisicing. Nostrud ea deserunt ad minim eu ex ullamco ex aliquip amet do dolor. Elit id cupidatat laborum pariatur nulla eu elit ut amet duis do enim.\r\n",
```

5.46 kB

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```
1  [
2    '{{repeat(120)}}',
3  ]
4  {
5    _id: '{{objectId()}}',
6    index: '{{index()}}',
7    age: '{{integer(20, 40)}}',
8    name: '{{firstName()}} {{surname()}}',
9    gender: '{{gender()}}',
10   company: '{{company().toUpperCase()}}',
11   email: '{{email()}}',
12   address: '{{integer(100, 999)}} {{street()}} {{city()}} {{state()}} {{integer(100, 10000)}}',
13 }
```

```
1  [
2    {
3      "_id": "65085c3eeaf1d8670fb43c78",
4      "index": 0,
5      "age": 29,
6      "name": "Norma Ortega",
7      "gender": "female",
8      "company": "ZILLANET",
9      "email": "normaortega@zillanet.com",
10     "address": "536 Heath Place, Stonybrook, Indiana, 2526"
11   },
12   {
13     "_id": "65085c3e4383240d0deb94a0",
14     "index": 1,
15     "age": 25,
16     "name": "Martina Lane",
17     "gender": "female",
18     "company": "QUILCH",
19     "email": "martinalane@quilch.com",
20     "address": "917 Barbey Street, Brenton, Federated States Of Micronesia, 9475"
21   },
22   {
23     "_id": "65085c3e5bbc42dedc3d9ad3",
24     "index": 2,
25     "age": 31,
26     "name": "Francis Cherry",
27     "gender": "female",
28     "company": "MUSIX",
29     "email": "francischerry@musix.com",
30     "address": "199 Rutledge Street, Blanco, Guam, 6788"
31   },
32   {
33     "_id": "65085c3ee184ce72617aeb77",
34     "index": 3,
35     "age": 20,
36     "name": "Rosetta Levy",
37     "gender": "female",
38     "company": "SHOPABOUT",
39     "email": "rosettalevy@shopabout.com",
40     "address": "232 Miami Court, Albrightsville, Oregon, 9733"
41   },
42   {
43     "_id": "65085c3e873b4f6f6a99df2b",
44     "index": 4,
45     "age": 23,
46     "name": "Santana Rollins",
47     "gender": "male",
48     "company": "SNOWPOKE",
49     "email": "santanarollins@snowpoke.com",
50     "address": "392 Mermaid Avenue, Waterview, Kansas, 137"
51   },
52   {
53     "_id": "65085c3ea7cb2ae22f62832b",
54     "index": 5,
55     "age": 32,
56     "name": "Henson Mcdaniel",
57     "gender": "male",
58     "company": "CEPRENE",
59     "email": "hensonmcdaniel@ceprene.com",
60     "address": "852 Herkimer Street, Yonah, Colorado, 3000"
61   },
62   {
63     "_id": "65085c3e0a6e57cec27a419e",
64     "index": 6,
65     "age": 36,
66     "name": "Barber Page",
67     "gender": "male",
```

24.31 kB

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THANK YOU.



TACK.



MERCI.

ANY QUESTIONS?